



THE ANNUAL NATIONAL LEVEL ROBOTICS COMPETITION, PCCOE&R, PUNE

ROBO RASHTRA '27

RISE OF ROBONATION, BHARAT LEADS THE WAY

YANTRAUTSAV

RULEBOOK

BUILD WHAT COMES NEXT





WELCOME NOTE

Welcome to Yantrautsav, an exciting event calling all aspiring innovators and young minds to showcase their creativity, passion, and robotics skills!

- Open to all skill levels, whether you're a beginner or an experienced creator in innovation.
- Golden opportunity to showcase innovative ideas and robotic creations.
- Open to participants of all age groups, ensuring diversity and exciting experiences.
- Demonstrate the technology behind your creation and explain its working.
- Solve real-world problems or introduce new concepts using this platform to share unique ideas.
- Impress the audience with your ingenuity for a chance to win exciting prizes!



EVENT OVERVIEW

In Yantrautsav, participants are invited to showcase their ingenuity and engineering skills by bringing a fully functional robotics-related model for the exhibition.

This challenge is designed to inspire creativity and innovation, allowing participants to demonstrate their understanding of robotics principles in a practical setting.

The competition is divided into two categories:

- Junior League for younger or less experienced participants.
- Senior League for advanced participants.

Both leagues will be evaluated on design, functionality and presentation, providing a platform for all skill levels to shine.



EVENT CATEGORIES

Category	Junior Category (Students of HSC or below)	Senior Category (Undergraduates and above)
Eligibility	Students of HSC (12th) or below	Undergraduates and above
Objective	Exhibit self-made robots, explain their working and unique features.	Showcase working robots or prototypes with a focus on innovation and problem-solving.
Prize - 1st	₹15,000	₹25,000
Prize - 2nd	₹10,000	₹20,000
Prize - 3rd	₹5,000	₹15,000
Special Note	Working robots gain extra points.	Fully operational bots gain extra points.



PRIZE POOL

Senior Category

Total Prize Pool: ₹60,000

- 1st Prize: ₹25,000
- 2nd Prize: ₹20,000
- 3rd Prize: ₹15,000

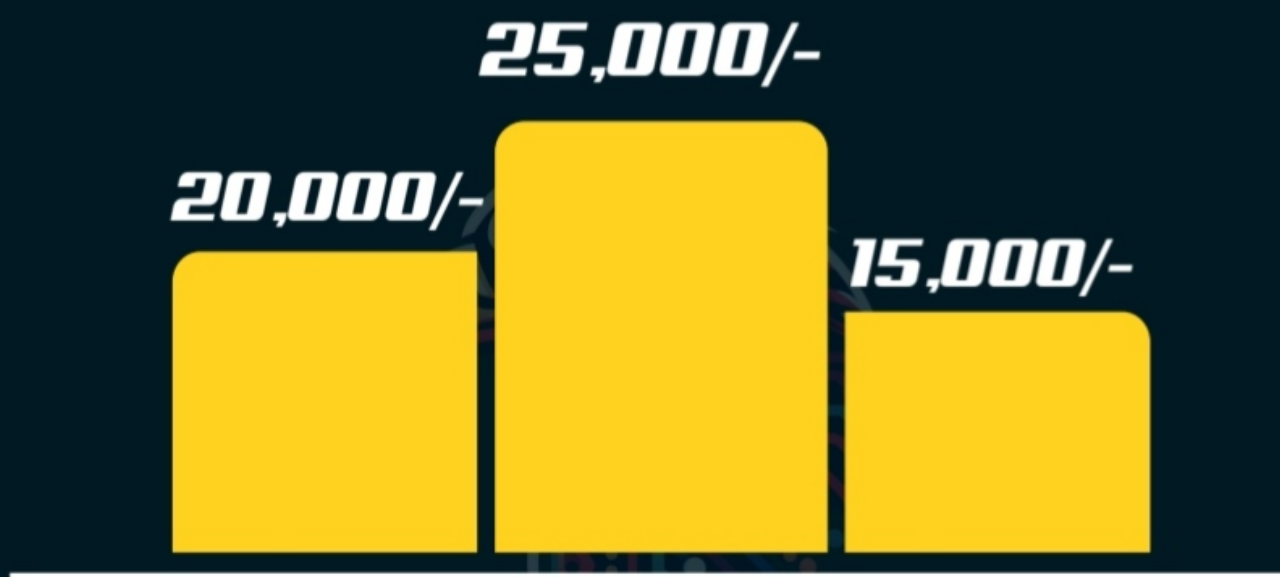
Junior Category

Total Prize Pool: ₹30,000

- 1st Prize: ₹15,000
- 2nd Prize: ₹10,000
- 3rd Prize: ₹5,000



SENIOR CATEGORY



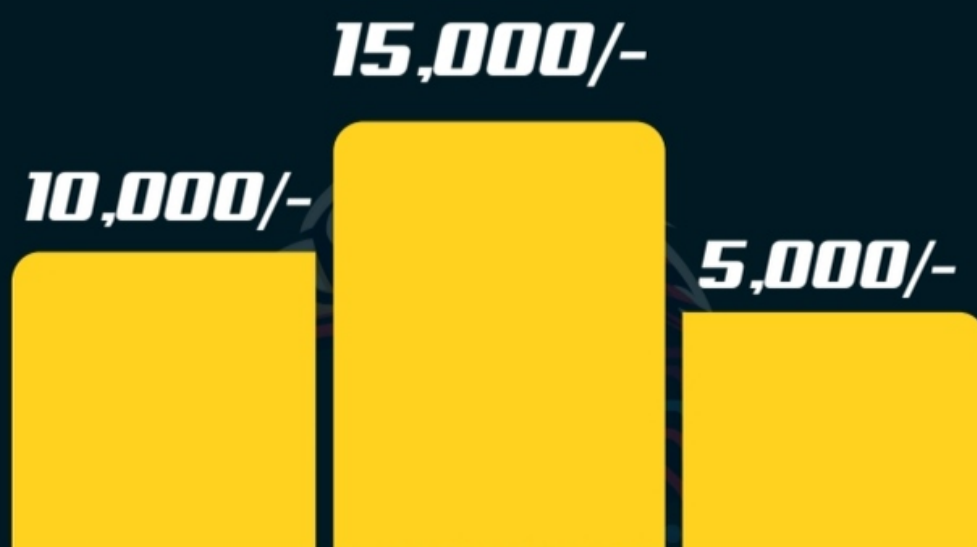
PRIZE POOL

60,000/-





JUNIOR CATEGORY



PRIZE POOL

30,000/-





ROBOTICS APPLICATION DOMAINS

The following domains are applicable to both Junior and Senior categories, but are not limited to:

4.1 Sustainable Energy

Wind, hydroelectric, solar power, etc.

4.2 Task-Performing Autonomous Projects

Autonomous robots, autonomous drones, etc.



ROBOTICS APPLICATION DOMAINS



Subcategories:

- Outdoor Autonomous Projects
- Indoor Autonomous Projects, e.g. smart home projects
- Other examples:
 - Arduino-based Snake Robot controlled using Android app
 - Wall climbing robot
 - Robotic vehicle controlled by hand gesture

4.3 Task-Performing Remote-Controlled Projects

Projects involving remotely controlled robotic systems.

4.4 Agricultural Projects

Grass cutter, Agribot, agricultural tasks.



ROBOTICS APPLICATION DOMAINS



4.5 Robotic Arms

Examples:

- Simple robotic arm
- Flex sensor arm
- Algorithm-based arm
- Android-controlled arm
- Hand-motion-controlled arm

4.6 Military Application Projects

Projects related to military applications.

4.7 Humanoid and Social Projects

Projects involving humanoid and social robotics.



ROBOTICS APPLICATION DOMAINS



4.8 Mechanical Education Projects

Examples:

- Emergency braking system in four-wheeler
- 8-leg spider robot using Theo Jansen linkage

4.9 Aquatic Projects

Examples:

- Solar lake pool cleaning drone
- Oil skimmer RC boat

4.10 Medical Field-Related Projects

Projects related to medical applications.



ROBOTICS APPLICATION DOMAINS



4.11 Swarm Projects

Projects involving swarm robotics.

4.12 Educational Robotic Projects

Projects intended for educational robotics.

4.13 Entertainment and Art Projects

Projects involving robotics for entertainment and artistic applications.



MARKING STRUCTURE

Total: 100 Marks

Criteria	Marks
1. Working Status	25
2. Innovation & Originality	20
3. Technical Feasibility & Implementation	20
4. Presentation Skills	15
5. Problem Statement & Relevance	10
6. Q&A; and Judges' Interaction	10
Total	100



TEAM SPECIFICATIONS

- Each team may consist of 1-4 members.
- Teams must register their bot and members prior to the event.



NOTE

Organizers reserve the right to modify or amend the competition rules at any time.

Participants will be notified of any changes.

The decision made by judges is final and cannot be contested.